

	the p.d. across the fixed resistor increases.
8(a)	They will hit the ground at the same time as they have the same initial vertical velocity and acceleration.
8(b)(i)	Using $v^2 = u^2 + 2as$ $v^2 = 0 + 2(9.81)(1.2)$ $v = 4.85 \text{ m s}^{-1}$
8(b)(ii)	Thermal energy transferred is equal to the loss in G.P.E. $E = (0.057)(9.81)(1.2 - 0.95)$ $= 0.14 \text{ J}$
8(b)(iii)	Proportion of energy remaining after one bounce $= \frac{0.95}{1.2} = 0.792$ Initial energy, $E_o = (0.057)(9.81)(1.2)$. After one bounce, $E_1 = 0.792E_o$. After 2 nd bounce, $E_2 = 0.792E_1 = (0.792)^2 E_o$. After nth bounce, $E_n = (0.792)^n E_o = mg(0.35)$. $(0.792)^n (0.057)(9.81)(1.2) = (0.057)(9.81)(0.35)$ $n = 5.3$ Minimum number of bounce = 6.

8(c)(i)	Horizontal velocity is constant = 3.60 m s^{-1} . Vertical velocity is same as ball A = 4.85 m s^{-1}
8(c)(ii)	Applying conservation of momentum, $(0.057)(3.6) = (0.057 + 0.154)v$ $v = 0.973 \text{ m s}^{-1}$
8(c)(iii)	As the trolley moves, there is work done against frictional forces and thus it loses kinetic energy and will slow down.
8(d)(i)	The trolley or the wall can be padded with cushion or foam to increase the time of taken for the trolley to slow down.
8(d)(ii)	Applying Newton's 2 nd law, $F\Delta t = \Delta p$ $(0.65)\Delta t = (0.057 + 0.154)(0.25 - 0)$ $\Delta t = 0.081 \text{ s}$